



EcoPile Overview Package

Introduction to EcoPile

The EcoPile, developed by Shoreline Plastics, LLC, is a revolutionary PVC/fiberglass composite piling specifically designed solely around the marine construction industry. It provides an environmentally friendly alternative to traditional treated timber pilings, which can leach harmful chemicals into waterways. The EcoPile is a composite piling created with the marine market in mind, delivering unmatched longevity and value for dock construction and other waterfront applications. It is engineered to offer a lifetime of durability with minimal maintenance.

Benefits of EcoPile

- **Environmentally Friendly:** EcoPile eliminates the need for treated wood, protecting aquatic ecosystems from harmful chemicals.
- **Exceptional Durability:** Resistant to rot, decay, marine borers, and UV degradation.
- **Cost-Effective:** Long-term savings due to minimal maintenance and unparalleled lifespan.
- **High Strength-to-Weight Ratio:** Offers significant load-bearing capacity and resistance to lateral forces while being lightweight for ease of handling.
- **Tailored Design:** Unique spiral threads for enhanced stability and superior skin friction

Installation Guidelines

Proper Pile Driving Techniques

1. Compatibility with Driving Equipment:

The EcoPile is compatible with nearly all types of pile-driving equipment, including:

- Drop hammers
- Diesel hammers
- Vibratory drivers

Recommendation: Vibratory drivers are optimal for overcoming the high skin friction generated by EcoPile's internal webbing, particularly for deep installations.

2. Cushion Blocks:

For larger equipment, Shoreline Plastics recommends using a cushion block, such as ¾-inch plywood sheets, wedged inside the hammer to prevent damage.

3. Avoid Side-Gripping Vibratory Hammers:

These can exert excessive clamping pressure, potentially damaging the EcoPile. Clamping force regulation may reduce damage but also compromises driving energy efficiency.

Unique Spiral Thread Design



The EcoPile features spiral “corkscrew” grooves on its lower 20%, improving stability by:

- Enhancing soil retention
- Increasing skin friction for superior bearing capacity

Installation Tip: Ensure the grooves are driven at least 1 foot below the mudline to avoid exposure and maximize strength under extreme load conditions.

Recommended Piling Depths

EcoPile provides up to five times the surface area of a wood pole of similar diameter. Once driven a few feet, soil compaction within the webbing generates substantial skin friction, often meeting load-bearing requirements at shallower depths than timber pilings. For larger projects, consult a geotechnical engineer to determine optimal depths.

Tool Selection

Lifting and Strapping

1. Strap Selection:

Use a well gripping choker strap or specialty straps, such as [Lift-It straps](#) with rubber fingers, for enhanced grip on the EcoPile’s smooth surface. Avoid chains directly on the EcoPile to prevent slipping or scratching. Wrap chains with a rubber hose if necessary.

2. Handling Recommendations:

EcoPile’s smooth outer shell can be slippery when wet; always prioritize grip strength to ensure safe handling during installation.

Cutting the EcoPile

1. Recommended Cutting Method:

The best way to cut the EcoPile is using a **sawzall (reciprocating saw)** with a blade featuring **8–12 teeth per inch**.

- **Preferred Blade Type:** Carbide-tipped blades are recommended to maximize lifespan and maintain cutting efficiency.
- **Advantages:** Clean, precise cuts without damaging the material.

2. Chainsaws: (not recommended) While chainsaws can be used to cut the EcoPile, they are not the preferred method for the following reasons:

- Chainsaw blades dull quickly when cutting composite materials.
- Cuts can leave the top of the pole chipped and uneven, similar to cutting vinyl seawalls.
- **Avoid in Freezing Temperatures:** The EcoPile becomes more brittle in cold weather, increasing the risk of chipping or cracking.

Drilling the EcoPile

1. Standard Drilling Option:

A standard **auger bit** works well for most applications.

2. Best Drilling Option:

For optimal results, use a **“nail killer” style auger bit** with the following features:

- **Carbide-Tipped Head:** Ensures longevity and effective drilling through composite material.
- **Less Aggressive Cutting Head Angles:** Reduces the risk of the drill kicking back when penetrating through the backside of the pole.

By selecting the right tools, contractors can ensure efficient installation, minimize wear on equipment, and maintain the structural integrity of the EcoPile. Always prioritize carbide-tipped options for cutting and drilling to maximize precision and tool lifespan.

Bolted Connections

1. Structural Connections:

- Use through bolts or all-thread rods with nuts and washers on both ends.
- Avoid lag bolts, as the semi-hollow core does not provide sufficient material for threading.

2. Temporary Fasteners:

Standard deck screws can hold boards in place temporarily before installing permanent through bolts.

3. Crushing Resistance:

EcoPile's internal webbing offers exceptional resistance to crushing. Torque connections until snug without over-tightening, especially on notched poles, to maintain structural integrity.

Design Considerations

1. Flexibility vs. Strength:

While the EcoPile may flex slightly more than new timber, it can absorb higher lateral loads without breaking. Flexibility ensures structural resilience during extreme weather conditions.

2. Connection Integration:

Tie EcoPile poles into the overall structure wherever possible to reduce sway and increase stability. For example:

- Use all-thread rods to connect free-standing poles to nearby stringers or substructures.

3. **Cross Bracing:**

In areas with strong currents or harsh conditions, cross bracing may be necessary, just as with timber poles.

Lift Selection

1. **Avoid Heavy Side Loads:**

Avoid lift designs with heavy cantilevered loads, such as beam-less or low-profile boat lifts with independent motors mounted directly to poles.

2. **Reinforcement for High Loads:**

For heavy side loads, consider filling the pole with concrete for additional support. Use curved washers (minimum 2.5-inch diameter) to distribute loads over a larger surface area.

Why Choose EcoPile?

EcoPile represents the future of marine construction, offering unmatched environmental and structural benefits. From docks to piers, this innovative solution provides lasting value and peace of mind for contractors and property owners alike.

Safety Data Sheet (SDS)

[\(click to download\)](#)

Section 1. Identification

Product identifier: EcoPile

PVC/fiberglass composite, encapsulated in PVC cap stock, structural marine piling

Trade names and Synonyms: N/A

Recommended Uses: Marine structural piling for docks, piers, etc.

Restrictions on Use: None

Company Identification & SDS Contact:

Shoreline Plastics, LLC

Attention: Ryan Boylan-South, J.D.

7167 Old Kings Road

Jacksonville, Florida 32219

Telephone (General): 904-696-2981

Emergency Telephone: 904-696-2981

Section 2. Hazard Identification

This product is not classified as hazardous under the U.S. Occupational Safety and Health Administration Hazard Communication Standard (HCS), 29 C.F.R. 1910.1200. PVC piling, a finished product, is not a hazardous chemical under normal conditions of use and is categorized as an article. While an SDS is not required under the HCS, this document is being provided as a courtesy to our customers.

NFPA Rating:

Health: 0



Fire: 0

Reactivity: 0

Special: None

Section 3. Composition/Information on Ingredients

Composition and Form manufactured rigid solid tubes of various dimensions for plumbing and water distribution as described in Section 1.

Polyvinyl Chloride

CAS: 9002-86-2

Weight: 75%-25%

Fiberglass

CAS: 659997-17-3

Weight: 25%-75%

Section 4. First-Aid Measures

EYE CONTACT: No effects anticipated under normal conditions of use. Particles generated by mechanical cutting, grinding or sanding can cause mechanical irritation. Immediately flush with large amounts of water, occasionally lifting the upper and lower eyelids to remove particles. Consult a physician if pain or irritation persists.

SKIN CONTACT: No effects anticipated under normal conditions of use.

INHALATION: The product is not expected to present an inhalation hazard unless mechanically chipped or pulverized or if melted during fire. If dust or fumes are inhaled, remove to fresh air.

SWALLOWING: No adverse health effects expected from ingestion.

Description of the most important symptoms or effects, and any symptoms that are acute or delayed: none known.

Recommendations for immediate medical care and special treatment needed, when necessary:

none known.

Section 5. Fire-Fighting Measures

PVC PILING DOES NOT PRESENT A FIRE OR EXPLOSION HAZARD UNDER NORMAL CONDITIONS OF USE. ALTHOUGH PVC PILING WILL NOT SUPPORT COMBUSTION, IT WILL BURN, RELEASING HYDROGEN CHLORIDE GAS, DETECTABLE BY ITS PUNGENT ODOR.

SPECIAL FIRE FIGHTING PROCEDURES: Evacuate enclosed areas. Confined areas require self-contained breathing apparatus. Extinguish using dry chemical, carbon dioxide, foam or water fog or spray equipment.

Section 6. Accidental Release Measures

Minor Spills Collect products and bundle or secure safely. If necessary, isolate area to prevent damage to/destruction of products by vehicles etc. Broken parts may be sharp and eye protection and gloves are recommended.

Major Spills Isolate area as necessary to prevent further damage. Collect products and bundle or secure safely. Broken product and parts may have sharp edges and eye protection and gloves are recommended.

Section 7. Handling and Storage

PVC pilings present no inhalation, ingestion or contact hazards. Cutting and grinding PVC piling may release nuisance dust particles which are non-toxic. Care should be taken to avoid inhaling dust. Use any methods that keep dust to a minimum. General storage procedures are acceptable.

Section 8. Exposure Controls/Personal Protection

Exposure Controls No exposure controls are necessary as products are inert and all ingredients are encapsulated within the polymer matrix and are believed to present no hazard under conditions of normal use and good occupational work practice.

Personal Protection:

Eyes: Glasses are recommended when handling piling and especially when working pilings mechanically, e.g. sawing, etc.

Hands/Feet: Safety footwear and gloves.

Engineering Controls: Appropriate controls for safe working when handling and mechanically working e.g. sawing, etc.

Section 9. Physical and Chemical Properties

- Appearance (physical state, color, etc.): Round piling with interior structural webbing. Color of piling varies upon request (e.g. black, white, grey, brown).
 - Upper/lower flammability or explosive limits: Not applicable
 - Odor: Not available
 - Vapor pressure: Not applicable
 - Odor threshold: Not available
 - Vapor density: Not applicable
 - pH: Not applicable
 - Relative density: Not Applicable
 - Melting point/freezing point: Not Applicable
 - Solubility(ies): Insoluble in water
 - Initial boiling point and boiling range: Not applicable
 - Flash point: Not applicable
 - Evaporation rate: Not applicable
 - Flammability (solid, gas): Will burn in contact with flame
 - Partition coefficient: n-octanol/water: Not applicable
 - Auto-ignition temperature: Not available
 - Decomposition temperature: Not available
 - Viscosity: Not applicable
-

Section 10. Stability and Reactivity

Reactivity

- Stable at normal temperatures and pressures.

Chemical stability

- Stable under normal storage conditions.

Other

- HAZARDOUS DECOMPOSITION PRODUCTS: CO, CO₂, hydrogen chloride and small amounts of benzene and aromatic and aliphatic hydrocarbons could be released in a fire.
 - Incompatible materials: Do not store with oxidizing agents.
-

Section 11. Toxicological Information

No toxicological data were found for this product. The effects reported are those anticipated based on the components of this product.

POTENTIAL ROUTES OF EXPOSURE: Exposure under ordinary use conditions to hazardous chemicals is not anticipated. Inhalation of dust from mechanical cutting, sanding, or grinding may occur.

SIGNS, SYMPTOMS, AND TOXIC EFFECTS OF OVEREXPOSURE: Exposure to high concentrations of dust of this product may cause irritation of the respiratory tract with cough, difficulty breathing, dryness of the throat, or eye irritation.

ANIMAL TOXICITY DATA: No data found

REPRODUCTIVE EFFECTS: No data were found regarding reproductive effects in humans or animals of any component of this product.

MUTAGENICITY DATA: No mutagenicity data were found for any component of this product.

DESIGNATION AS POTENTIAL CARCINOGEN: IARC designates PVC homopolymer as Group 3, "not classifiable as to its carcinogenicity in humans," and titanium dioxide as Group 2B, "Possibly carcinogenic to humans".

MEDICAL CONDITIONS AGGRAVATED BY EXPOSURE: No data were found regarding this issue.

INTERACTIONS WITH CHEMICALS THAT ENHANCE TOXICITY: No data were found regarding this issue.

Section 12. Ecological Information

No Data were found regarding adverse ecological impacts of this product.

Section 13. Disposal Considerations

Recycle where possible. Refer to state/territory environmental protection agency/authority.
Normally suitable for disposal as general waste land fill.

Section 14. Transportation Information

Not Regulated.

Section 15. Regulatory Information

This plastic PVC piling is an article and, therefore, exposure to titanium dioxide is unlikely because the substances are inextricably bound in the plastic matrix. It is unlikely that titanium dioxide will contribute to workplace exposures under normal conditions of use. While there is a possibility that mechanical cutting, sanding or grinding the product could produce respirable particles, it is not clear that the titanium dioxide would be found as unbound particles of respirable size. Users must determine if respirable particles are produced in their operations. If so, then the appropriate Prop 65 warning language is as follows:

California's Proposition 65

 **WARNING:** This product can expose you to chemicals including titanium dioxide, which are known to the State of California to cause cancer. For more information go to www.P65Warnings.ca.gov.

Please note that Prop 65 only requires the identification of one chemical per endpoint in the warning.

Section 16. Other Information

Users Responsibility

A bulletin such as this cannot be expected to cover all possible individual situations. As the user has the responsibility to provide a safe workplace, all aspects of an individual operation should be examined to determine if, or where, precautions, in addition to those described herein, should be passed on to your customers or employees. Shoreline Plastics, LLC, must rely on the user to utilize the information we have supplied to develop work practice guidelines and employee instructional programs for their individual operations.

Disclaimer of Liability

As the conditions or methods of users are beyond our control, we do not assume any responsibility and expressly disclaim any liability for any use of this material. Information contained herein is believed to be true and accurate, but all statements or suggestions are made without warranty, expressed or implied, regarding accuracy of the information, the hazards connected with the use of the material, are the responsibility of the user. The information used in this letter was compiled from other MSDS letters with similar products.

Engineering Report ([Click to download](#))



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Finite Element Analysis Report

for the

PVC Pilings

for

Shoreline Plastics, LLC

300 Alton Box Rd W
Jacksonville, FL 32218

Revision 1: October 22, 2019
Report Date: September 17, 2019

Prepared by:

John Loesel

Senior Consultant

Reviewed by:

Jeremy Orr, P.E.

Principal Engineer - Manager

ATS Job Number: 321753 Purchase Order Number: 51425

Professional Engineers and Scientists: Design • Consulting • Testing • Inspection

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Background and Finite Element Analysis Results

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Finite Element Analysis Plots

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Attachments

Client Supplied Dimensions and Material Specifications.



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Background and Finite Element Analysis Results

Background

This report contains the results of the finite element analysis (FEA) performed on one design of a PVC piling in four (4) different sizes, which were provided by Shoreline Plastics LLC (client). The client requested the analysis to compare the deflection of various piling designs when subjected to a lateral load of 200lbs, as well as a buckling analysis to determine an estimated buckling load limit. Per the requested scope of work, a linear static analysis was performed. Frequency, transient, thermal, or other types of loading were not considered in the analysis. The maximum resultant displacements and theoretical buckling limit for the four sizes were estimated with the FEA analysis and are presented below.

Revision 1 Note: The client supplied dimensions for design EP8 which have been added to the report.

Finite Element Analysis - Setup

The finite element models consisting of solid elements were constructed from the client provided cross section drawings. Five(5) sizes of one (1) piling design cross section were submitted for analysis and were named EP8, EP10, EP12, EP14, EP16 in reference to their outer diameter. Refer to the charts below for dimensions of each size. The models were based on the client provided dimensions with direction from the client. ATS did not modify or simplify the designs for the analysis. Loading was assumed to be steady state, and did not include analysis of any acceleration, deceleration, or transient effects. The material was a client specified Rigid PVC with Stranded Fiberglass Blend and the client provided the relevant material properties which are located in the attachments.

A representative 10' section of piling was modeled for each cross section. The base of the piling was fixed (to simulate/approximate the bottom of the pipe being buried). A 200lb lateral load was applied to the very top of the pilings. A linear static analysis was then performed using the specified loads and restraints to determine a lateral displacement for comparison.

Component	Material	Yield Strength (MPa)	UTS (psi)	Youngs Modulus (MPa)	Shear Modulus (MPa)	Poisson's Ratio
Pilings (5x)	Rigid PVC w/ Stranded Fiberglass Blend (Client Spec)	103	N/A	9,650	3,644	0.324

The following loads were considered in the analysis:

Load	Value	Notes
Lateral Load	200 lbs	Load applied at the top face of the pilings.



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Background and Finite Element Analysis Results

Finite Element Analysis - Results

The following results were obtained using the above linear static analysis, restraints, and loads. Solidworks 2017 was also used to calculate the area moment of inertia. Follow-up hand calculations using cantilever beam formulas matched the results of the FEA study to within 0.005". The estimated stresses for all beams were less than 11% of the stated yield strength of 103 MPa and all simulations were determined to be within the linear elastic range and non-linear analysis was not performed (as well as being outside the requested scope of work).

Piling Design	O.D. (in)	O.D. Thick. (in)	Int. Thick. (in)	Area Moment Of Inertia (lowest) in ⁴	FEA Estimated Deflection @10ft Length (in)	Beam Calculation Deflection (in)
8	8.000	0.280	0.175	72.9	1.135	1.129
10	10.000	0.300	0.175	152.9	0.543	0.538
12	12.000	0.340	0.230	315.4	0.264	0.261
14	14.000	0.350	0.250	533.0	0.157	0.154
16	16.000	0.360	0.300	873.2	0.097	0.094

The design with the least amount of deflection was determined to be EP16 as seen in the above chart. ATS did not perform factor of safety, stress, thermal, or frequency analysis on these parts. This analysis does not constitute an engineering review or evaluation for use of the piling or overall system design. The analysis was limited to determining the least amount of lateral deflection to the subjected load.

Maximum displacement plots for each design are located in "Finite Element Analysis Plots" that immediately follows.


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Background and Finite Element Analysis Results
Finite Element Analysis - Buckling Results

Solidworks 2017 was used to determine a theoretical Buckling Load Factor (BLF) when the client specified length of piling (which differed for each size) was subjected to a vertical 10,000 lb load. Per the client, 30% of the span length is buried. The buried portion of the piping was fixed in place and vertical load was applied to the very top of the pilings. Refer to the attached images for a visual representation of the buckling FEA setup.

The theoretical estimated buckling load factor and buckling limit are listed below. It is important to note that the numbers shown below are estimates based on simple linear buckling analysis. The approach that was followed for this analysis (using a linear elastic buckling model without bifurcation analysis) should be heavily scrutinized and likely results in non-conservative values. This is due to the assumed perfect geometries as compared to reality where geometric nonlinearities exist. Discretization errors also effect the results. In general, such an approach should be used as a first screening in a complete buckling study which was outside the requested scope of work.

Piling Design	O.D. (in)	O.D. Thick. (in)	Int. Thick. (in)	Overall Length Of Column (ft)	Buried Length (30%) (ft)	Unsupporte d Column Length (ft)	Buckling Load Factor at 10,000 lbs	FEA Estimated Buckling Limit (lbs)
8	8.000	0.280	0.175	20.0	6.0	14.0	1.25	12,463
10	10.000	0.300	0.175	20.0	6.0	14.0	2.03	20,300
12	12.000	0.340	0.230	30.0	9.0	21.0	1.73	17,300
14	14.000	0.350	0.250	40.0	12.0	28.0	1.62	16,237
16	16.000	0.360	0.300	50.0	15.0	35.0	1.70	17,024



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Finite Element Analysis Plots

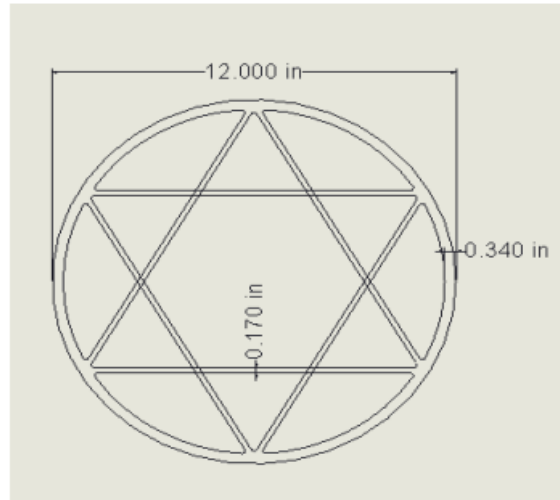


Photo 1: Typical client supplied piling cross section (sizes and thickness vary per size).

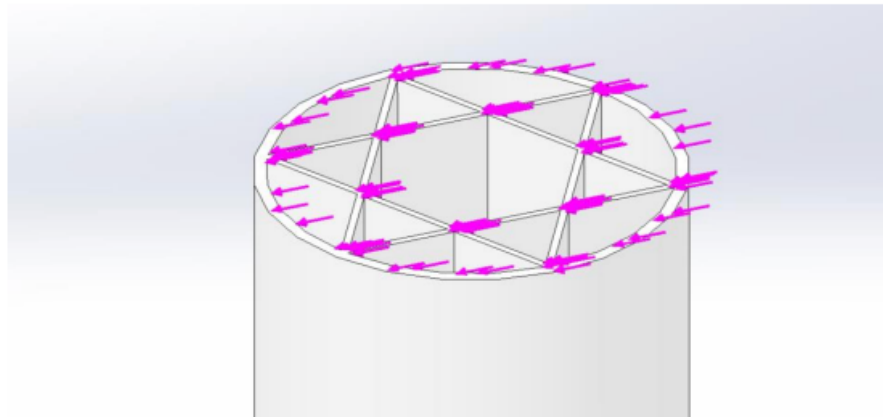


Photo 2: Typical view of 200lb lateral load applied to the top face of the pilings.



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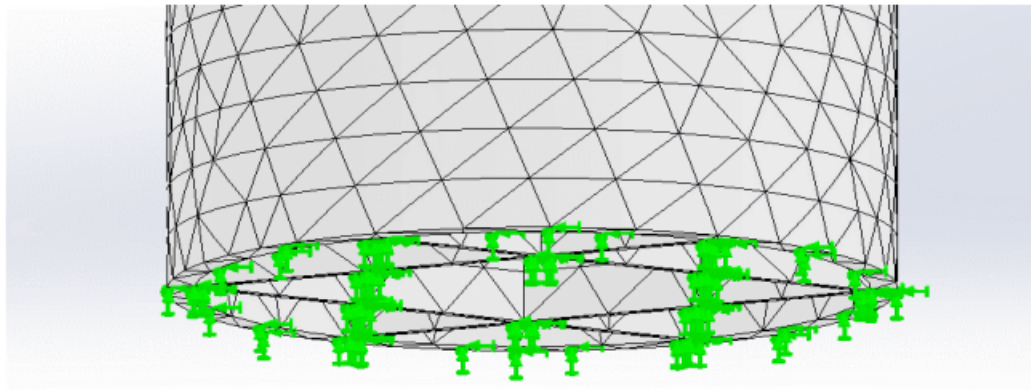


Photo 3: Typical view of the fixed restraint to the bottom of the pilings.

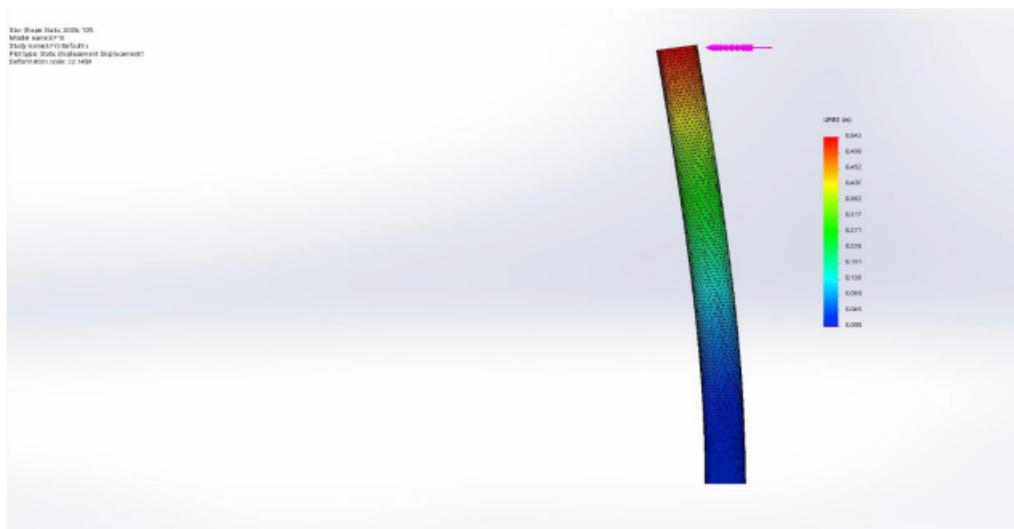


Photo 4: "EP10" resulting displacement plot. Maximum displacement=0.543".



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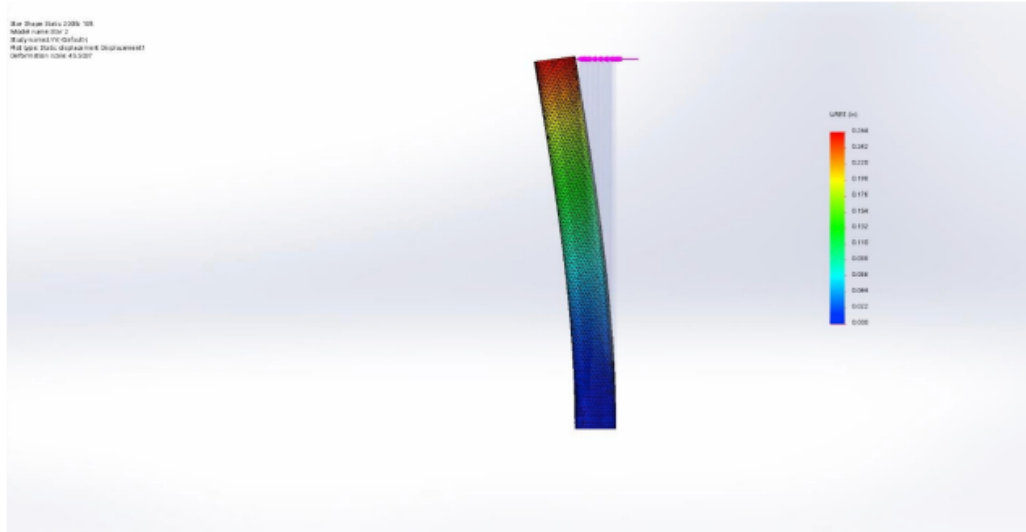


Photo 5: "EP12" resulting displacement plot. Maximum displacement=0.264".

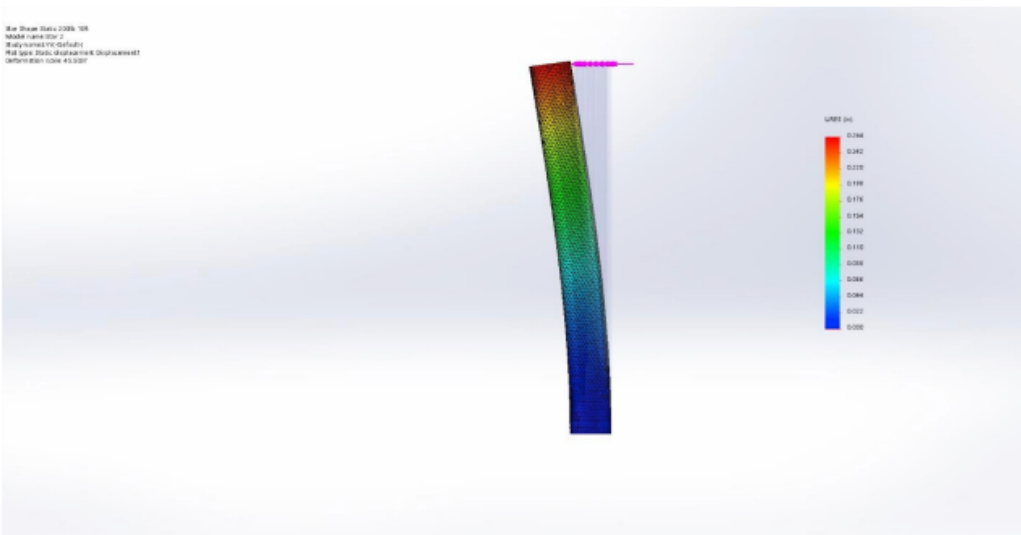


Photo 6: "EP14" resultant displacement plot. Maximum displacement=0.157".



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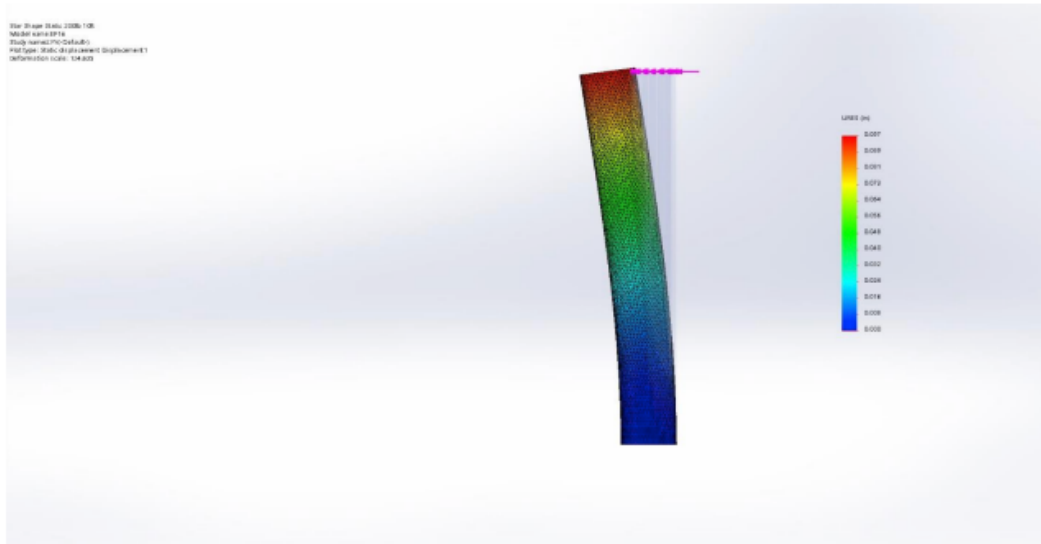


Photo 7: "EP16" resultant displacement plot. Maximum displacement=0.097".

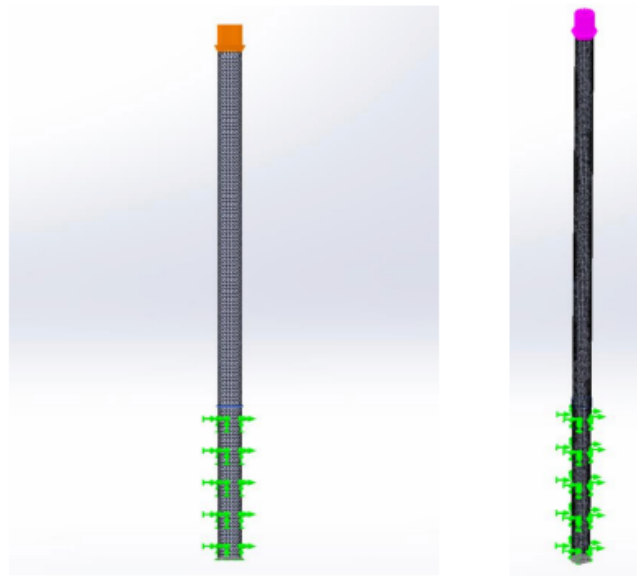


Photo 8: Typical FEA simulation set-up for buckling analysis. Note the "buried" portion of the piling has been fixed and a vertical load applied at the top of the piling.



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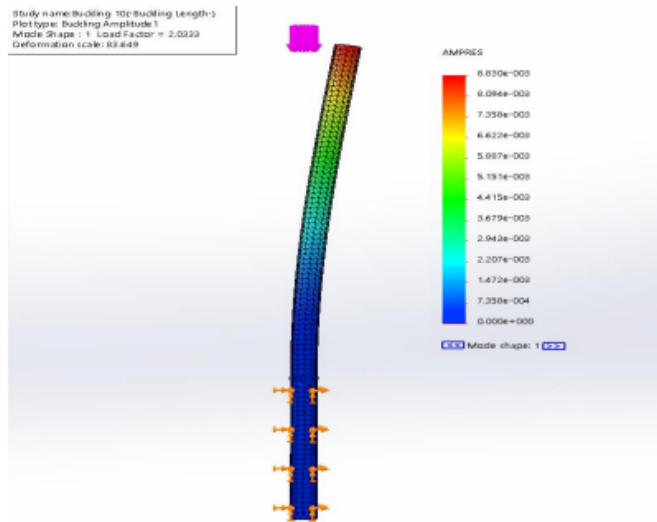


Photo 9: "EP10" under 10,000lb load. Resultant Buckling Load Factor = 2.03.

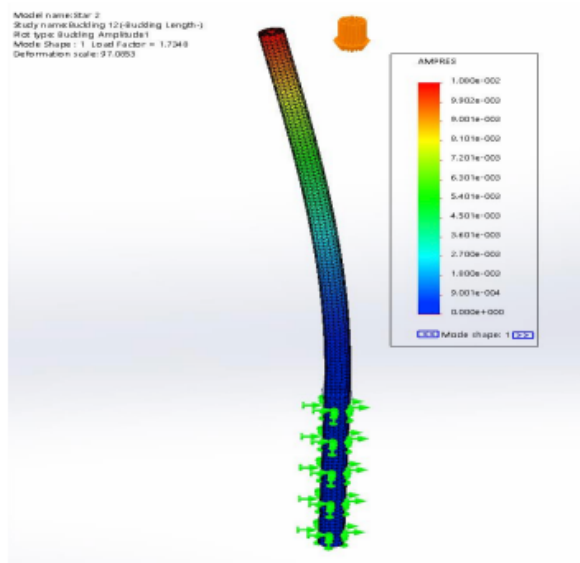


Photo 10: "EP12" under 10,000lb load. Resultant Buckling Load Factor = 1.73.



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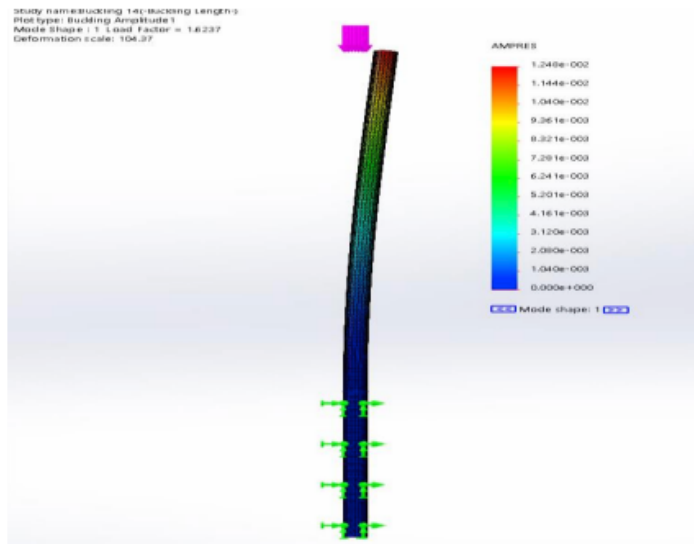


Photo 11: "EP14" under 10,000lb load. Resultant Buckling Load Factor = 1.62.

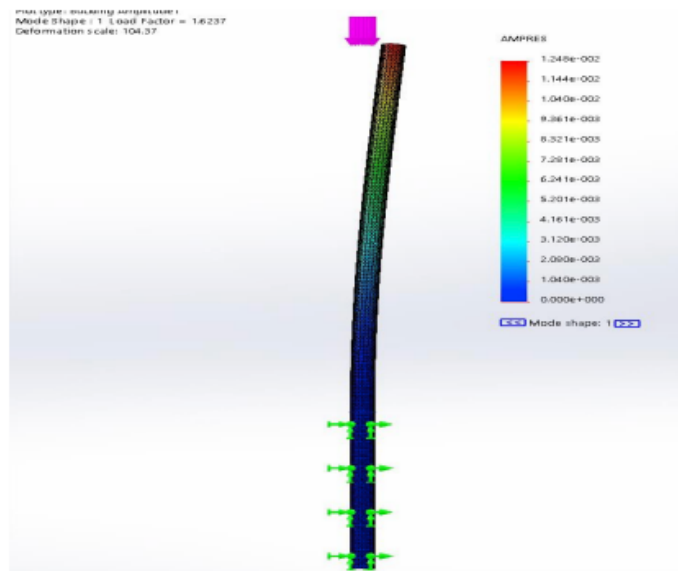


Photo 12: "EP16" under 10,000lb load. Resultant Buckling Load Factor = 1.70.



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Finite Element Analysis Plots

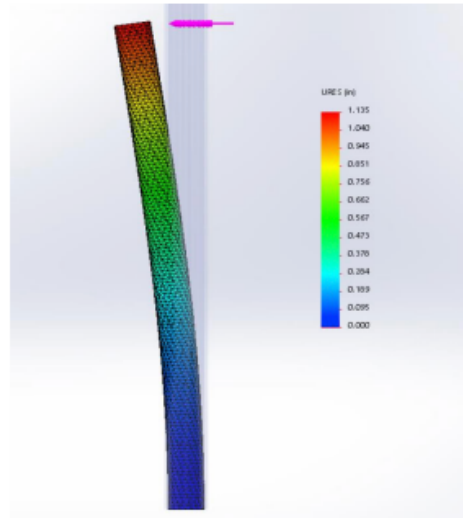


Photo 13: "EP8" resulting displacement plot. Maximum displacement=1.135".

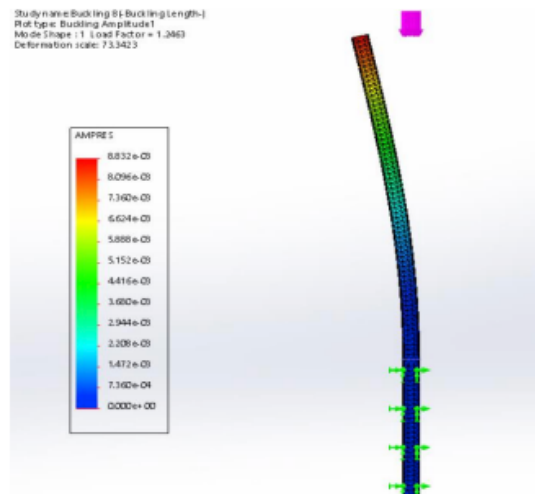


Photo 14: "EP8" under 10,000lb load. Resultant Buckling Load Factor = 1.25.



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Client Supplied Dimensions and Material Specifications

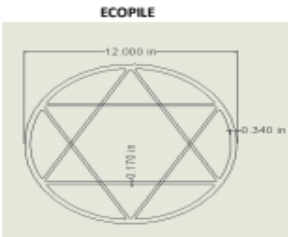
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ECOPILE MATERIAL PERFORMANCE DATA

SHORELINEPLASTICS.COM

FIBER REINFORCED PVC COMPOSITE MARINE PILE

ECOPILE	ECOPILE DIAMETER		ECOPILE YIELD STRENGTH		ECOPILE MODULUS OF ELASTICITY		ECOPILE BEARING CAPACITY		ECOPILE FLEXURAL MODULUS		ECOPILE WEIGHT	
	INCH	mm	LB/IN ²	Mpa	PSI	Mpa	PSI	Mpa	PSI	Mpa	LB/FT	KG/M
EP8	8.00	203.20									10.64	1.47
EP10	10.00	254.00									12.69	1.75
EP12	12.00	304.80	15,000	103.00	1.40E + 6 PSI	9,650	208,000	1,434	1.35E+6 PSI	9310	16.33	2.26
EP14	14.00	355.60									17.87	2.47
EP17	17.00	431.80									21.75	3.01



Product Data Sheet

ECOPILE			
Manufacturer	Shoreline Plastics LLC		
Product	PVC/FIBERGLASS COMPOSITE MARINE PILING		
MATERIAL			
Raw Material	Polyvinyl chloride (PVC),		
Extruded Material	Ridged PVC compounded with Stranded Fiberglass blend		
Co-Extruded Material	UV protected PVC Capstock		
Molecular Weight (resin)	65	K value	
Relative Viscosity	2.15	dL/gm	
Inherent Viscosity	0.91	dL/gm	
Relative Density	1.47		
Cell Classification	N/A		
MECHANICAL PROPERTIES:			
Impact Strength: Notched Izod	110 J/m	23 deg C	ASTM D-256A
Impact strength: Unnotched Izod	270 J/m	23 deg C	ASTM D-256
Tear strength:	750-1000 lb ft/in ASTM D-624		
Hardness:	88	Shore D durometer	
Coefficient of Friction:	0.3 μ s	PVC to PVC	
Tensile Strength @ Break: %	2%	ASTM-D 638	
Tensile Stress @ Yield:	103	103 MPa	ASTM D 638
Tensile Modulus:	9650	MPa	ASTM D638
Flexural Modulus:	9310	MPa	ASTM D790
THERMAL PROPERTIES:			
Softening Point:	120 C		
Max continuous service Temp:	65 C		
FIRE PERFORMANCE:			
Flammability: Oxygen index	0.45	ASTM D 2863	
Ignitability index	2	AS 1530 (tested as a tube)	
Smoke produced index	8	AS 1530 (tested as a tube)	
Flame spread index	0	Does not support combustion	

ELECTRIC PROPERTIES:	
Dielectric strength (Breakdown)	14-20 kV/mm
Volume resistivity	10^{16} Ohm.cm (60% RH)
Surface resistivity	$10^{13} - 10^{14}$ Ohm
Dielectric constant	3.4-3.6 at 25° C (60Hz)
Dissipation factor	0.015-0.020 at 20°C
NOTES:	
Ultra Violet Weathering: ECOPILE maintains the same level of UV protection as Vinyl siding and Vinyl windows with an exposure rating in excess of 50 years. ECOPILE is a two layered product, the inner layer is a PVC/Fiberglass composite, the outer layer is an impervious, heavy, PVC coating that is an accepted medium for the containment of potable water. PVC is immune to Microbiological and Macrobiological attack. Testing conducted by Technical University of Darmstadt Germany	
Sources	
DATA SHEET NO.	

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Appendix —

Buckling Limits

Estimated Allowable Buckling Limits (lbs) with a Safety Factor of 2.5

Length of Piling Exposed Above Mudline (Assumes Solid Hold at Mudline - Add a Few Extra Feet if Working in Softer Soil Conditions) -Unit Pounds					
Piling Diameter	10 ft	12 ft	14 ft	16 ft	18 ft
8" Piling	9,771	6,785	4,985	3,818	3,018
10" Piling	15,915	11,052	8,120	6,219	4,916
12" HD Piling	54,620	37,931	27,868	21,305	16,834
16" Piling	83,418	57,929	42,560	32,544	25,712
	22 ft	24 ft	26 ft	28 ft	30 ft
8" Piling	2,019	1,696	1,449	1,255	1,103
10" Piling	3,288	2,763	2,360	2,044	1,797
12" HD Piling	11,269	9,469	8,081	7,007	6,136
16" Piling	17,235	14,482	12,360	10,717	9,394
	32 ft	34 ft	36 ft	38 ft	40 ft
8" Piling	977	872	783	708	644
10" Piling	1,595	1,421	1,276	1,153	1,048
12" HD Piling	5,439	4,852	4,364	3,951	3,597
16" Piling	8,324	7,427	6,679	6,051	5,514

Buckling capacity based on a freestanding piling, when bolted into structure, load capacity increases considerably. Please speak to local engineer to verify your loading capacities



Appendix —

Bolted Connection Performance Summary

Overview

The following section summarizes the allowable loading data, torque recommendations, and design considerations for bolted connections through the EcoPile composite piling system. The data were developed using the mechanical properties of the PVC/fiberglass composite used in the 12" EcoPile and modeled under the most conservative assumptions, including *worst-case bolt orientation* relative to the internal web structure (see Photo A).

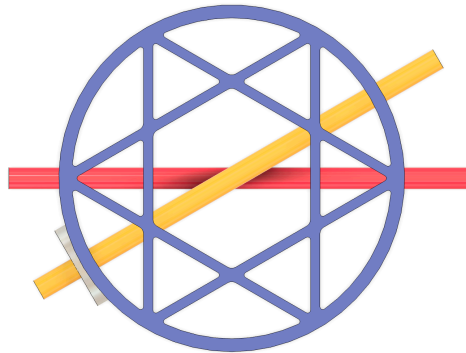
These guidelines provide general design information and should be verified by a licensed engineer for project-specific conditions.

1. Vertical Load Capacity (Bolt Bearing / Wallowing Resistance)

Allowable downward loads for bolts passing through the EcoPile wall are shown below. Values assume minimum contact area between bolt and internal webbing (worst-case “unsupported” orientation).

Bolt Diameter	Allowable Load (lbs)	Allowable Torque (in-lb)	Allowable Torque (ft-lb)
1/2"	3,260	19,550	1,630
3/4"	4,890	29,350	2,446

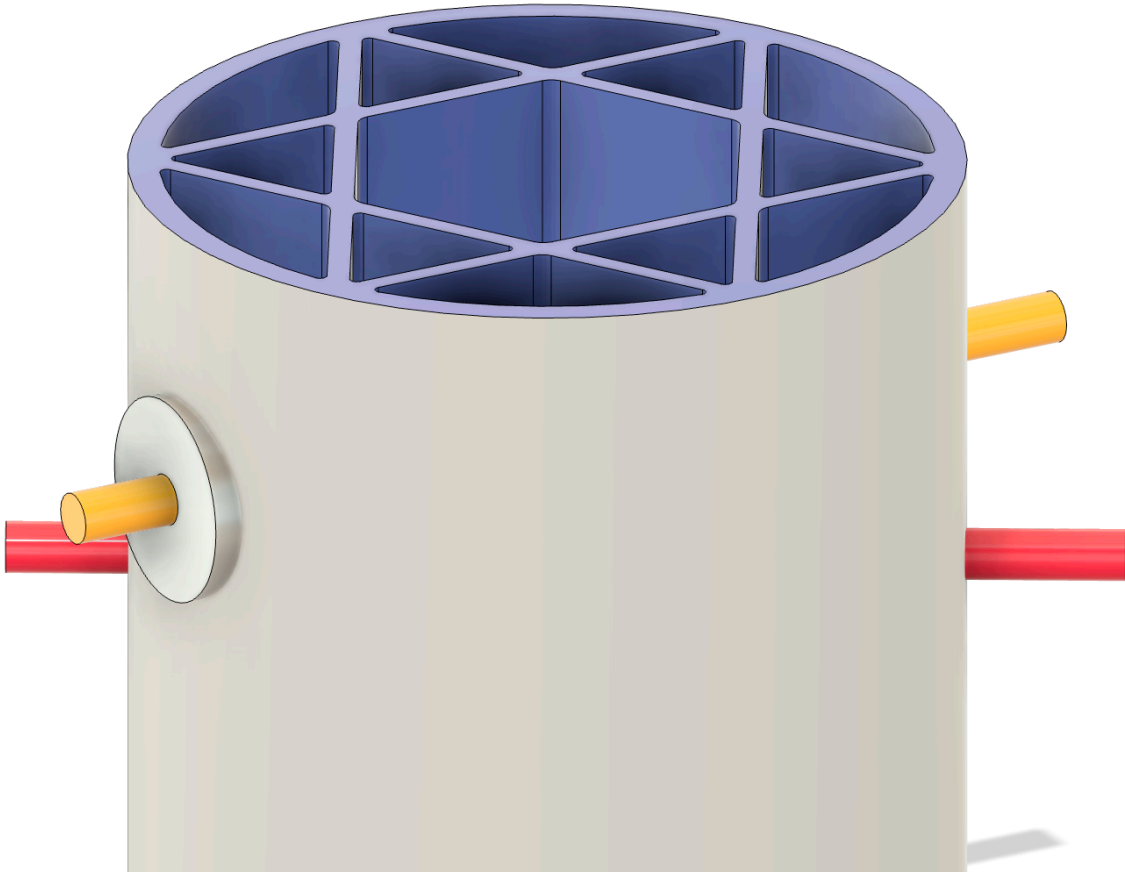
These values correspond to the least favorable alignment shown in Photo A, where the bolt passes between webs rather than through them.



2. Crushing and Clamp Load Resistance

Crushing resistance beneath the washer varies depending on both washer size and internal web orientation (see Photo B).

To evaluate local compression and deflection, calculations were performed using curved washers of 1.5", 2", and 3" diameters under both *rib-backed (supported)* and *unsupported* bolt paths.





Washer Diameter	Orientation	Condition	Clamp Force (lbf)	Recommended Torque (ft-lb, K = 0.20)	Torque Range (K = 0.18–0.25)
1.5" Curved Washer	Rib-Backed (Supported)	No Deflection	4,847	61	55–76
	Rib-Backed	Crack Risk / Failure	9,694	121	109–151
	Unsupported	No Deflection	3,231	40	36–51
	Unsupported	Crack Risk / Failure	6,463	81	73–101
2" Curved Washer	Rib-Backed	No Deflection	10,116	126	114–158
	Rib-Backed	Crack Risk / Failure	20,232	253	228–316
	Unsupported	No Deflection	6,744	84	76–105
	Unsupported	Crack Risk / Failure	13,488	169	152–211
3" Curved Washer	Rib-Backed	No Deflection	25,169	315	283–393
	Rib-Backed	Crack Risk / Failure	50,339	629	566–787
	Unsupported	No Deflection	16,780	210	189–262
	Unsupported	Crack Risk / Failure	33,559	420	378–524

Note: For typical field installations, 3" curved washers distribute load most effectively, minimizing local deformation. Apply a 25–50% safety factor to ensure conservative design margins.

3. Recommended Torque and Fastening Practice

Field trials have confirmed that 3/4" stainless through-bolts can be torqued up to approximately 1,000–1,100 ft-lb using a professional-grade impact driver, without damaging the pile when proper washers are used.

Recommended best practices:

- Tighten until snug, not over-compressed.
 - Always use curved washers $\geq 3"$ for notched or high-load locations.
 - Use through-bolts or all-thread rods with nuts and washers on both sides; avoid lag bolts.
-

4. Design Considerations

- **Orientation Effect (see Photo A):**
Bolts intersecting internal webs experience greater bearing strength and reduced deflection compared to unsupported orientations. Position fasteners to intersect at least one internal web when possible.
- **Load Distribution (see Photo B):**
Larger washers or outer reinforcement rings distribute clamp force evenly, reducing local stresses on the composite shell.
- **Cap Plate Systems (see Photo C – optional):**
For roof beam or lift connections, a top-mounted cap plate system transfers load vertically through the top face of the piling, using through-bolts primarily for uplift restraint.
- **Safety Margin:**
Actual connection strength depends on orientation, washer size, and reinforcement. Always confirm with a qualified engineer for project-specific conditions.

5. Typical Applications

- Deck or roof beam fastening using 1/2"–3/4" stainless through-bolts.
- Boat-lift bracket attachment using large aluminum or steel tabs for load spreading.
- Retaining pole and cap-plate assemblies subject to combined uplift and lateral loads.

Summary

The EcoPile's internal web structure provides excellent resistance to both wallowing and crushing under bolted connections.

When installed using proper washers and torque control, EcoPile connections achieve exceptional structural integrity and long-term reliability across a wide range of marine applications.



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Links-

[Safety Data Sheet \(SDS\)](#)

[Engineering Report](#)

[Dansco Engineering Code Approval](#)

[Google drive \(several photos and videos of the EcoPile being worked with,
that are not listed on the website.](#)